BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Secondary Storage

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why might an operating system use virtual memory?

2. Which is a benefit of a high-level language?
   A) Never needs translation
   B) Easier for humans to read
   C) Runs directly on every CPU
   D) Uses no programming rules

3. Convert binary 10010110 to hexadecimal, using four-bit groups.

4. Use decimal units. A file is 4 MB. Give its size in KB and bytes.

Today's objectives

Identify magnetic, optical and solid-state storage technologies. Compare storage choices using capacity, cost, speed and durability.

Key vocabulary - Read, cover, recall

Magnetic storage
Stores bits as magnetic patterns.

Optical storage
Uses lasers to read data from discs.

Solid-state storage
Non-volatile storage with no moving parts.

Cost per GB
Price per unit of stored capacity.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Magnetic
An HDD stores data magnetically on rotating platters; it has moving mechanical parts.

Optical
Discs such as DVDs are read by lasers interacting with a reflective surface.

Solid state
SSDs and USB flash drives use non-volatile flash memory with no moving parts.

Choose for context
Compare capacity, cost per GB, speed, portability, reliability and durability; justify trade-offs.

I Do - Worked model

A photographer carries equipment between locations. Explain one advantage and one disadvantage of choosing an SSD instead of an HDD. [4 marks]

- Identify the use case
- The drive will travel often and hold large files.
- SSD has no moving parts and is more resistant to impacts; access can also be faster.
- An SSD may cost more per GB than an HDD; justify one trade-off. [4]

We Do - Guided practice

1. Choose storage for a large, low-cost archive that is rarely moved. Explain your choice. [4]

2. A school sends files on DVDs. Identify the technology and explain a limitation. [3]

3. Explain why an SSD might be suitable for a moving laptop. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Calculate how many complete 400 MB files fit on a 2,000 MB drive.

```
capacityMB = 2000
fileMB = 400
files = ...
print(files)
```

You Do - Independent exam questions

Q2. Name the storage technology used by (i) DVD, (ii) HDD and (iii) USB flash drive. [3]

Q3. Explain one advantage and one disadvantage of magnetic storage compared with solid-state storage. [4]

Q4. A school needs portable, reusable storage for transporting files. Justify choosing a USB flash drive. [3]

Exit Ticket - True or False?

An algorithm only gives reliable results when its inputs are handled as specified.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. When there is not enough free RAM, secondary storage is used as virtual memory.

2. B - Easier for humans to read

3. 96: 1001 0110.

4. (a) 4,000 KB and 4,000,000 bytes.

I Do - Explain the model

Identify the use case

The drive will travel often and hold large files.

SSD has no moving parts and is more resistant to impacts; access can also be faster.

An SSD may cost more per GB than an HDD; justify one trade-off. [4]

We Do - Model answers

1. Magnetic HDD is suitable [1] for high capacities [1] at relatively low cost per GB [1]; slower access may be acceptable for infrequent use [1].

2. Optical storage [1]; the disc is read using a laser [1] and may have limited capacity compared with modern SSDs or HDDs [1].

3. No moving parts [1] improves resistance to shocks [1], and fast read/write times [1] suit regular use.

Code First - Answer

```
capacityMB = 2000
fileMB = 400
files = capacityMB DIV fileMB
print(files)
```

Expected output: 5

You Do - Self-mark

Q2: (i) Optical [1]; (ii) magnetic [1]; (iii) solid-state/flash [1].

Q3: Magnetic HDD often costs less per GB [1], advantageous for large capacity [1]; it has moving parts [1], increasing vulnerability to physical shock [1].

Q4: Solid-state [1], compact and portable [1], reusable with no moving parts [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.